

EE5750 Quantum Error Correction

Homework 6

Due: 10 Sep 2026

As part of the Homework assessment, provide solutions to only the questions marked as (***Homework***). The rest of the questions are practice problems.

1. (***Homework***) (5 marks) *'Spooky action-at-a-distance'*

Recall that entanglement between two registers A, B can be created only when a joint physical operation (such as a unitary gate) can be applied to A and B . For instance, A and B can be brought together in a lab, a joint unitary can be applied, and afterwards A and B can be separated (no matter how far in space) while still maintaining their entangled state.

Suppose *qubits* A and B are initially prepared in state $|0\rangle$ each.

- (a) Find a unitary gate that acts on AB to create the entangled pair

$$\frac{1}{\sqrt{2}} (|01\rangle + |10\rangle).$$

- (b) Suppose we measure the observable Z on qubit A (this is a local measurement), what are the possible post-measurement states of AB and with what probabilities do they occur?

Remark: Neither of the two post-measurement states of AB is equal to the pre-measurement state. These two outcomes and their probabilities remain valid even if A and B are separated in space (after creating the entanglement). We observe that a physical intervention done locally at A instantaneously affects the global state of AB , i.e., a local action creates an immediate global update even if the two qubits are located far away from each other. This is what Einstein called 'spooky action-at-a-distance'.

2. (***Homework***) (10 marks) *Creating entanglement using measurement*

You showed in the previous question that entanglement can be created by doing a unitary operation on the composite system AB . We can also create entanglement using measurements, although we might not have control over which entangled state we end up with.

Assume that we start with independently prepared qubits A and B in states $|0\rangle$ each. Design a projective measurement with 4 possible outcomes such that each of the post-measurement states of AB is an entangled state.

You must provide the following in your solutions:

- (a) an observable corresponding to your projective measurement, and steps to show that the projection operators correspond to the distinct eigenvalues of your observable.
- (b) proof that each of the 4 possible post-measurement states is actually an entangled state.

3. (***Homework***) (10 marks) *Copying orthogonal states*

The no-cloning theorem (which we discussed very briefly during Lec 12) says that we cannot build a quantum circuit that can create two independent copies (output of the circuit) of an arbitrary state (input to the circuit). In classical computing 'copy-and-paste' is of course allowed; this does not

contradict the quantum no-cloning theorem — you *can* clone an input state if it is restricted to a pre-determined orthonormal set. In the classical setup this happens to be computational basis states $|0\rangle$, $|1\rangle$, etc.

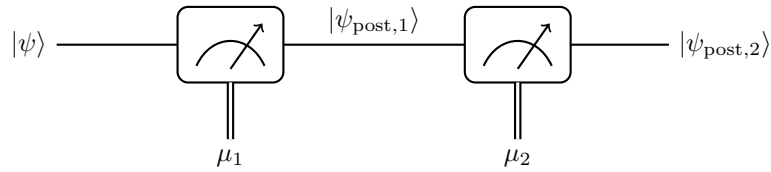
Suppose the qubit A is the input to a quantum computer, and qubit B is a register *present within* the computer (initialized to state $|0\rangle$). The output of the computer is AB .

- Build a unitary gate for the quantum computer (to act on input AB) to clone the state of A into the state of B (so that AB is in a product state) provided that the state of A is restricted to $|0\rangle$ or $|1\rangle$ only.
- What is the output of your circuit if the input register A is in state $|+\rangle$? Is B a clone of A ?
- Repeat part (a) for input restricted to $|+\rangle$ and $|-\rangle$.

4. **(*Homework*)** (10 marks) *Repeated measurements*

Consider a projective measurement on $\mathbb{C}^d$ described by the observable $M = \sum_{\ell=1}^L \lambda_\ell P_\ell$, with distinct real numbers $\lambda_1, \dots, \lambda_L$. The operators $P_1, \dots, P_L$ are projections ($P_\ell^2 = P_\ell^\dagger = P_\ell$) that are complete, $\sum_{\ell=1}^L P_\ell = I$, and mutually orthogonal, $P_i P_j = 0$ for $i \neq j$.

A register is prepared in a pure state $|\psi\rangle$, and the observable M is measured twice in succession, as shown below. Let μ_1 and $|\psi_{\text{post},1}\rangle$ be the classical outcome and post-measurement state of the first measurement, and let μ_2 and $|\psi_{\text{post},2}\rangle$ be the classical outcome and post-measurement state of the second measurement (which acts on $|\psi_{\text{post},1}\rangle$).



Assume that you (as an experimenter) can observe the classical outcomes of the measurements.

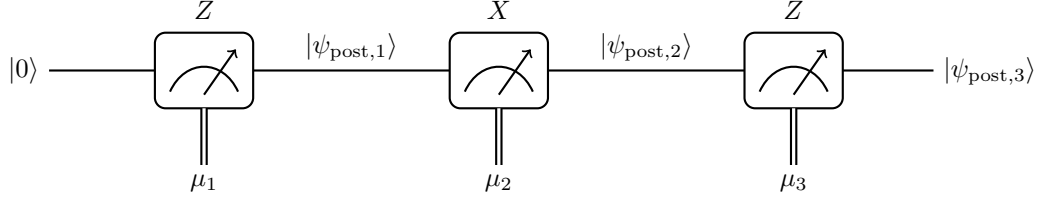
- Show that $\mu_2 = \mu_1$ with probability 1, and that $|\psi_{\text{post},2}\rangle = |\psi_{\text{post},1}\rangle$. Hence a projective measurement that is immediately repeated with the same observable changes neither the classical outcome nor the state of the register.
- Show that the same conclusion holds when the state fed into the first measurement is an arbitrary (possibly mixed) state ρ , that is, the two post-measurement states (possibly mixed states) are the same and the classical outcomes are the same.

Remark: This is the sense in which projective measurements are *repeatable*: the first measurement leaves the register in an eigenvector of M , and every subsequent measurement of M reads back the same outcome without disturbing the state any further.

5. **(*Homework*)** (5 marks) *On repeatability of measurements*

The repeatability observation holds only when the *same* observable is measured in succession. If these two measurements are interspersed with a measurement of a different observable, repeatability no longer holds.

The figure below shows three measurements performed in succession on a qubit prepared in the state $|0\rangle$; as in the previous question, μ_i denotes the classical outcome and $|\psi_{\text{post},i}\rangle$ the post-measurement state of the i -th measurement. The three measurements are of the observables Z, X, Z , respectively.



Argue that there is a non-zero probability that $\mu_1 \neq \mu_3$ and $|\psi_{\text{post},1}\rangle \neq |\psi_{\text{post},3}\rangle$ (although the same observable is being measured).

6. Deutsch's algorithm

This is an introduction to the simplest quantum algorithm that shows an advantage over classical computing. You can refer to Section 1.4.3 of Michael A. Nielsen and Isaac L. Chuang, *Quantum Computation and Quantum Information* (10th Anniversary Edition, Cambridge University Press, 2010), or Chapter 6 of Phillip Kaye, Raymond Laflamme, and Michele Mosca, *An Introduction to Quantum Computing* (Oxford University Press, 2007), for a detailed treatment.

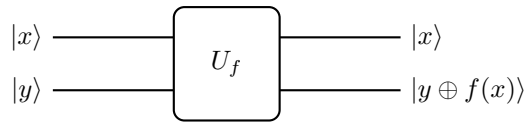
Querying a classical function quantumly.

Let $f : \{0, 1\} \rightarrow \{0, 1\}$ be a classical Boolean function. Think of f as a sealed device — an *oracle* — whose internal workings are hidden from us: we may *query* it by supplying an input bit x and reading off the output bit $f(x)$, but we cannot inspect how it computes.

To use such a device inside a quantum circuit it must be a *unitary* gate, and hence reversible. The map $x \mapsto f(x)$ on its own is in general *not* reversible — if f is constant, both inputs give the same output, so x cannot be recovered from $f(x)$. The standard remedy is to use two single-qubit registers, a *query register* and a *target register*, and to define a gate U_f on the computational basis by

$$U_f(|x\rangle \otimes |y\rangle) = |x\rangle \otimes |y \oplus f(x)\rangle, \quad x, y \in \{0, 1\},$$

where $\oplus$ denotes addition modulo 2; U_f is then extended linearly to superpositions of the four basis states $|00\rangle, |01\rangle, |10\rangle, |11\rangle$.



Setting the target register to $|0\rangle$ gives

$$U_f(|x\rangle \otimes |0\rangle) = |x\rangle \otimes |f(x)\rangle,$$

so a single application of U_f — one *query* — writes $f(x)$ into the target register while leaving the query register unchanged.

Thus, in the quantum scenario, querying a classical function is the same as applying the unitary U_f ; if you quantumly query the function n times in your algorithm, this means that you have applied the unitary gate U_f exactly n times in the circuit.

Query complexity.

In this black-box (oracle) model, an algorithm may access f only through the oracle — classically by supplying x and reading $f(x)$, quantumly by applying U_f — and its *query complexity* is the number of oracle calls it makes, in the worst case over the possible functions f . Every other operation (state preparation, ordinary gates, measurements) is counted as free, so the query complexity captures how many times the algorithm must interrogate f to complete its task. The question of interest is whether a quantum algorithm can decide a given property of f using strictly fewer queries than any classical algorithm; Deutsch's algorithm is the simplest example where it can.

The problem: constant or balanced.

There are exactly four functions $f : \{0, 1\} \rightarrow \{0, 1\}$. Two are *constant* — $f(0) = f(1) = 0$ and $f(0) = f(1) = 1$ — and the other two are *balanced*, meaning f takes the values 0 and 1 equally often; these are $f(x) = x$ and $f(x) = 1 \oplus x$. Thus f is constant when $f(0) = f(1)$ and balanced when $f(0) \neq f(1)$; equivalently, f is constant or balanced according as the single bit $f(0) \oplus f(1)$ equals 0 or 1.

Deutsch's problem.

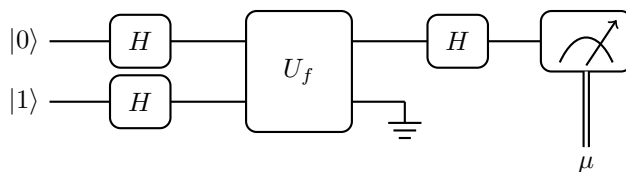
Given oracle access to an unknown f — as the black box $x \mapsto f(x)$ classically, or as the gate U_f quantumly — decide whether f is constant or balanced (equivalently, compute the bit $f(0) \oplus f(1)$), using as few queries as possible.

Classical query complexity.

A classical algorithm can learn about f only by evaluating it. One query returns a single value, say $f(0)$: this value is consistent both with the constant function equal to $f(0)$ everywhere and with the balanced function having $f(1) = 1 \oplus f(0)$, so a single query can never distinguish the two cases. Two queries always suffice: evaluate $f(0)$ and $f(1)$, and report “constant” if they are equal and “balanced” otherwise. Hence the classical query complexity of Deutsch’s problem is exactly 2.

Deutsch's algorithm.

This algorithm uses exactly one quantum query to solve this problem. The circuit for this algorithm is shown, where the query register is measured in the computational basis. You must trace the state through this circuit, step by step, and show that the value μ indicates whether the function f is constant or balanced.



For each of the four possible functions $f : \{0, 1\} \rightarrow \{0, 1\}$, find the classical outcome μ of the measurement. What is the value of μ when f is balanced, and what is it when f is constant?
